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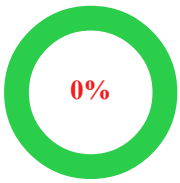

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ABSTRACT

Study Buddy is an integrated and intelligent academic collaboration platform designed to enhance peer-to-peer learning, content sharing, and real-time doubt resolution among students at Bannari Amman Institute of Technology (BIT). The system serves as a centralized digital hub that consolidates study materials, mentorship, discussion forums, and instant communication into a single platform, ensuring seamless academic support and enhanced learning experiences.

Built using React (Vite) for the frontend and Node.js with Express for the backend, with MySQL (Sequelize ORM) for data management, Study Buddy ensures a secure, scalable, and responsive architecture capable of supporting multiple concurrent users. The platform features a robust authentication system with email-password login to maintain user privacy and data security, ensuring

only authorized students and contributors can access or upload content.

A highlight of the system is its Real-Time Doubt-Solving Module, powered by Socket.io, which enables students to post academic queries and receive instant responses from seniors or mentors. This includes support for one-to-one chat sessions, group discussions, and live screen-sharing for guided doubt clarification. The Notes Management Module allows senior students to upload study materials such as PDFs, images, or slides, which can be categorized by department, semester, and subject. Juniors can rate and review these materials, ensuring quality and credibility while promoting an organized knowledge-sharing environment.

The platform also includes an Interactive Discussion Forum for collaborative learning, structured topic-based discussions, and thread-based replies. The Admin and Mentor Dashboard enables monitoring of uploaded content, active participants, and engagement metrics, allowing for effective mentorship, oversight, and analytics-driven insights. Designed with a responsive user interface using Tailwind CSS, the platform works flawlessly across desktops, laptops, and mobile devices, offering consistent accessibility and usability.

By integrating automated content management, intelligent real-time communication, and structured peer learning, Study Buddy promotes efficiency, transparency, and a collaborative academic culture. It empowers students to take vi

ownership of their learning, fosters mentorship between seniors and juniors, and provides a scalable, future-ready solution to support academic growth in modern digital campuses.

KEYWORDS: Academic Collaboration, Study Materials Management, Real-Time Doubt Resolution, Peer Mentorship, React (Vite), Node.js, Express, MySQL, Socket.io, Tailwind CSS, Student Engagement, Digital Learning Platform

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CHAPTER 1

INTRODUCTION

In today's technology-driven educational ecosystem, students face increasing challenges in accessing quality study materials, organizing academic resources, and communicating effectively with peers and seniors. The lack of a unified digital platform for sharing notes, clarifying doubts, and engaging in academic discussions leads to disconnection, redundancy, and inefficiency. Relying on fragmented systems such as messaging apps, email chains, and social media groups causes students to miss valuable academic opportunities and learning support.

To address these challenges, this project introduces Study Buddy – Smart Notes Sharing and Doubt-Solving Platform, an integrated and collaborative web application developed exclusively for students of Bannari Amman Institute of Technology (BIT). The system bridges the gap between senior and junior students by enabling the sharing of study materials, doubt clarification, peer discussions, and one-to-one mentoring sessions with real-time communication features. It provides a seamless digital space where students can upload notes, rate materials, ask queries, chat instantly, and even attend live screen-sharing doubt sessions with seniors.

By consolidating academic interaction, content sharing, and communication

into one digital platform, Study Buddy aims to promote collaborative learning, transparency, and digital empowerment among students. Built with React (Vite) for the frontend, Node.js + Express for the backend, and MySQL with Sequelize ORM for database management, the system ensures performance, scalability, and data integrity while offering a modern, user-friendly interface.

1.1 BACKGROUND OF THE WORK

The transformation of modern education through digital tools has improved accessibility but also introduced complexities in managing resources and communication. College students frequently rely on multiple platforms—such as Google Drive for notes, WhatsApp groups for doubts, and separate portals for

1

study materials—creating a disconnected academic experience. Faculty members and mentors also find it difficult to monitor student learning patterns or provide consistent academic guidance across such scattered mediums.

At Bannari Amman Institute of Technology, where collaborative peer learning is strongly encouraged, the lack of a dedicated internal system

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